Airline Management System

Standard Entity-Relationship (ER) Diagram & Attribute Specification

Model Scope & Classification: This report presents the **Standard Entity-Relationship (ER) Model** for the Airline Management System database. In strict accordance with classical ER modeling principles (non-enhanced, omitting subclassing / specialization / generalization hierarchies), this model explicitly details the **12 fundamental entities**, their **attributes** (Primary Keys, Foreign Keys, data types, and check constraints), and their **direct relational associations** normalized in Third Normal Form (3NF).

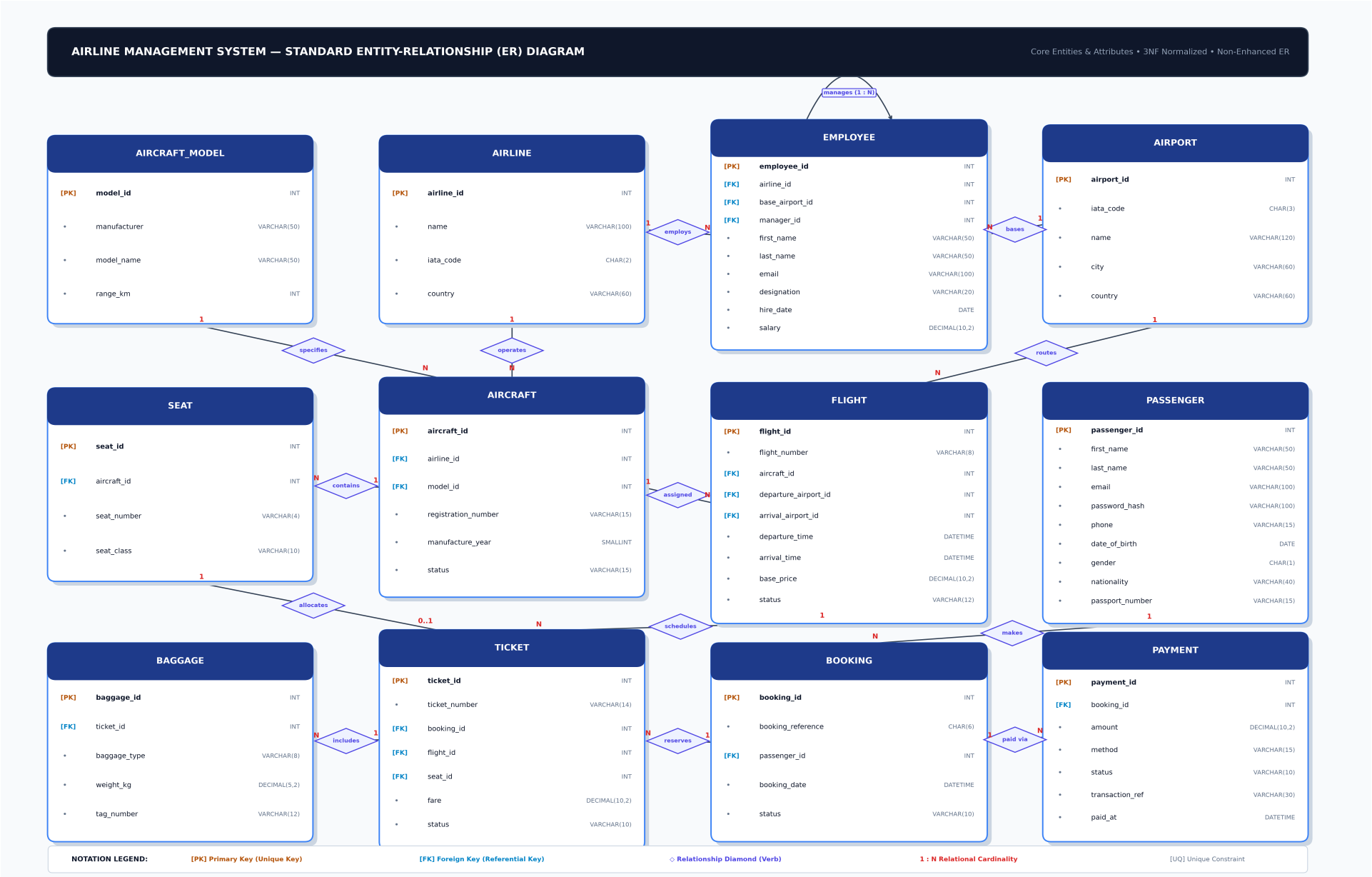
Total Entities: 12 **Direct Relationships:** 13 **Recursive Hierarchies:** 1 (Employee) **RDBMS Engine:** MySQL 8.0+ (InnoDB / 3NF)

System Entity Directory at a Glance

#	Entity Name	Primary Key [PK]	Foreign Keys [FK]	Core Business Function
1	**airline**	airline_id	None	Commercial airline operating companies
2	**airport**	airport_id	None	Commercial airport terminals and facilities
3	**aircraft_model**	model_id	None	Standard aircraft specifications and models
4	**aircraft**	aircraft_id	airline_id, model_id	Physical fleet airframes and registrations
5	**seat**	seat_id	aircraft_id	Cabin seat positions and class tiers
6	**flight**	flight_id	aircraft_id, dep_airport_id, arr_airport_id	Scheduled flight sectors between airports
7	**passenger**	passenger_id	None	Registered travelers and customer accounts
8	**booking**	booking_id	passenger_id	Flight reservations and PNR bookings
9	**ticket**	ticket_id	booking_id, flight_id, seat_id	Issued boarding coupons linking flight & seat
10	**payment**	payment_id	booking_id	Financial payment transactions and settlements
11	**baggage**	baggage_id	ticket_id	Checked and cabin baggage allowances
12	**employee**	employee_id	airline_id, base_airport_id, manager_id	Flight crew, ground crew, and managers

1. Graphical Entity-Relationship (ER) Diagram

Standard ER Diagram showcasing all 12 entities, their attribute lists with Primary Keys [PK] and Foreign Keys [FK], and direct relationship paths with cardinalities (1 : N, 1 : 1). Zero-crossing layout ensures optimal clarity.



2. Comprehensive Entity and Attribute Catalog

Complete technical catalog detailing each entity's attributes, data types, constraints, nullability, and relational linkages.

Entity: 1. AIRLINE — *Represents commercial airlines operating flights in the network.*

Relationships: • 1 : N with **aircraft** (owns/operates fleet)
• 1 : N with **employee** (employs airline workforce)

Attribute Name	Data Type	Key / Constraint	Description & Rules
airline_id	INT	[PK] Auto Increment	Unique surrogate primary key for the airline
name	VARCHAR(100)	UNIQUE, NOT NULL	Full registered commercial name (e.g., IndiGo, Air India)
iata_code	CHAR(2)	UNIQUE, NOT NULL	Two-letter standard IATA airline code (e.g., 6E, AI)
country	VARCHAR(60)	NOT NULL	Country of registration and headquarters

Entity: 2. AIRPORT — *Represents origin and destination airport facilities.*

Relationships: • 1 : N with **flight** as departure_airport_id
• 1 : N with **flight** as arrival_airport_id
• 1 : N with **employee** as base_airport_id

Attribute Name	Data Type	Key / Constraint	Description & Rules
airport_id	INT	[PK] Auto Increment	Unique surrogate primary key for the airport
iata_code	CHAR(3)	UNIQUE, NOT NULL	Standard three-letter IATA code (e.g., DEL, BOM, BLR)
name	VARCHAR(120)	NOT NULL	Official airport name (e.g., Indira Gandhi International Airport)
city	VARCHAR(60)	NOT NULL	City served by the airport
country	VARCHAR(60)	NOT NULL	Country where airport is located

Entity: 3. AIRCRAFT_MODEL — *Stores standardized aircraft manufacturing specifications and flight range.*

Relationships: • 1 : N with **aircraft** (categorizes physical fleet aircraft into standardized models)

Attribute Name	Data Type	Key / Constraint	Description & Rules
model_id	INT	**[PK] Auto Increment**	Unique surrogate primary key for the aircraft model
manufacturer	VARCHAR(50)	NOT NULL	Aircraft manufacturing company (e.g., Airbus, Boeing)
model_name	VARCHAR(50)	NOT NULL	Aircraft model designation (e.g., A320neo, B737-800, B787-9)
range_km	INT	NOT NULL, CHECK > 0	Maximum flight operational range in kilometers

Entity: 4. AIRCRAFT — *Represents individual physical aircraft airframes belonging to an airline fleet.*

Relationships: • N : 1 with **airline** (owned/operated by airline)
• N : 1 with **aircraft_model** (specifications reference)
• 1 : N with **seat** (contains passenger cabin seats)
• 1 : N with **flight** (assigned to scheduled flight sectors)

Attribute Name	Data Type	Key / Constraint	Description & Rules
aircraft_id	INT	**[PK] Auto Increment**	Unique surrogate primary key for the physical airframe
airline_id	INT	**[FK] Ref: airline**	Operating airline owning this aircraft
model_id	INT	**[FK] Ref: aircraft_model**	Aircraft model specifications reference
registration_number	VARCHAR(15)	UNIQUE, NOT NULL	Civil aviation tail number (e.g., VT-IZD, VT-EXO)
manufacture_year	SMALLINT	NOT NULL, CHECK 1990-2035	Year of aircraft manufacture and rollout
status	VARCHAR(15)	DEFAULT ACTIVE, CHECK	Fleet operational state: ACTIVE, MAINTENANCE, or RETIRED

Entity: 5. SEAT — *Represents individual cabin seat coordinates configured inside an aircraft.*

Relationships: • N : 1 with **aircraft** (configured inside physical aircraft)
• 1 : 0..1 with **ticket** (allocated to ticket for a scheduled flight sector)

Attribute Name	Data Type	Key / Constraint	Description & Rules
seat_id	INT	[PK] Auto Increment	Unique surrogate primary key for the seat configuration
aircraft_id	INT	[FK] Ref: aircraft	Aircraft airframe containing this seat
seat_number	VARCHAR(4)	NOT NULL	Seat coordinate inside cabin (e.g., 01A, 03B, 12F)
seat_class	VARCHAR(10)	NOT NULL, CHECK	Cabin service tier: ECONOMY or BUSINESS

Entity: 6. FLIGHT — *Represents scheduled commercial flight operations connecting two airports.*

Relationships: • N : 1 with **aircraft** (airframe deployed)
• N : 1 with **airport** (departure origin)
• N : 1 with **airport** (arrival destination)
• 1 : N with **ticket** (passenger tickets issued for flight sector)

Attribute Name	Data Type	Key / Constraint	Description & Rules
flight_id	INT	[PK] Auto Increment	Unique surrogate primary key for the scheduled flight sector
flight_number	VARCHAR(8)	NOT NULL	Flight code identifier (e.g., 6E-2041, AI-101)
aircraft_id	INT	[FK] Ref: aircraft	Physical aircraft deployed on this flight
departure_airport_id	INT	[FK] Ref: airport	Origin departure airport facility
arrival_airport_id	INT	[FK] Ref: airport	Destination arrival airport facility
departure_time	DATETIME	NOT NULL	Scheduled date and time of departure
arrival_time	DATETIME	NOT NULL, CHECK > dep	Scheduled date and time of arrival
base_price	DECIMAL(10,2)	NOT NULL, CHECK > 0	Base passenger airfare in local currency (INR)
status	VARCHAR(12)	DEFAULT SCHEDULED, CHECK	State: SCHEDULED, DELAYED, CANCELLED, DEPARTED, ARRIVED

Entity: 7. PASSENGER — *Stores customer identity profiles, travel documentation, and authentication data.*

Relationships: • 1 : N with **booking** (initiates passenger ticket reservations)

Attribute Name	Data Type	Key / Constraint	Description & Rules
passenger_id	INT	**[PK] Auto Increment**	Unique surrogate primary key for the passenger
first_name	VARCHAR(50)	NOT NULL	Passenger given first name
last_name	VARCHAR(50)	NOT NULL	Passenger family / surname
email	VARCHAR(100)	UNIQUE, NOT NULL	Registered login and notification email address
password_hash	VARCHAR(100)	NOT NULL	Cryptographic hash for authenticated portal access
phone	VARCHAR(15)	UNIQUE, NOT NULL	Primary contact telephone number
date_of_birth	DATE	NOT NULL	Date of birth for age verification and fare tiering
gender	CHAR(1)	CHECK M, F, O	Gender marker: M (Male), F (Female), O (Other)
nationality	VARCHAR(40)	DEFAULT Indian, NOT NULL	Country of legal citizenship
passport_number	VARCHAR(15)	UNIQUE, NULLABLE	International travel passport credential number

Entity: 8. BOOKING — *Represents passenger reservation transactions and booking PNR headers.*

Relationships: • N : 1 with **passenger** (made by registered customer)
• 1 : N with **ticket** (contains individual passenger flight tickets)
• 1 : N with **payment** (settled through financial payment transactions)

Attribute Name	Data Type	Key / Constraint	Description & Rules
booking_id	INT	**[PK] Auto Increment**	Unique surrogate primary key for the booking reservation
booking_reference	CHAR(6)	UNIQUE, NOT NULL	6-character alphanumeric PNR locator (e.g., PNR101, PNR102)
passenger_id	INT	**[FK] Ref: passenger**	Registered passenger who created the booking
booking_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp when booking reservation was submitted
status	VARCHAR(10)	DEFAULT PENDING, CHECK	Reservation lifecycle state: PENDING, CONFIRMED, CANCELLED

Entity: 9. TICKET — *Represents issued passenger flight coupons linking booking, flight, and allocated seat.*

Relationships: • N : 1 with **booking** (line item in passenger reservation)
• N : 1 with **flight** (scheduled flight sector reserved)
• N : 1 with **seat** (allocated cabin seat; set NULL on cancellation)
• 1 : N with **baggage** (associated registered baggage items)

Attribute Name	Data Type	Key / Constraint	Description & Rules
ticket_id	INT	**[PK] Auto Increment**	Unique surrogate primary key for the ticket
ticket_number	VARCHAR(14)	UNIQUE, NOT NULL	Standardized 14-character e-ticket serial number (e.g., ETKT-2026-0001)
booking_id	INT	**[FK] Ref: booking**	Associated parent booking reservation
flight_id	INT	**[FK] Ref: flight**	Reserved scheduled flight sector
seat_id	INT	**[FK] Ref: seat, NULLABLE**	Allocated aircraft seat (freed to NULL if ticket cancelled)
fare	DECIMAL(10,2)	NOT NULL, CHECK > 0	Final charged passenger fare for this sector
status	VARCHAR(10)	DEFAULT PENDING, CHECK	Coupon state: PENDING, ISSUED, or CANCELLED

Entity: 10. PAYMENT — *Records payment settlements and banking gateway transaction references.*

Relationships: • N : 1 with **booking** (settles financial balance for reservation booking)

Attribute Name	Data Type	Key / Constraint	Description & Rules
payment_id	INT	**[PK] Auto Increment**	Unique surrogate primary key for the payment record
booking_id	INT	**[FK] Ref: booking**	Associated reservation booking reference
amount	DECIMAL(10,2)	NOT NULL, CHECK > 0	Monetary settlement amount in INR
method	VARCHAR(15)	NOT NULL, CHECK	Payment channel: CARD, UPI, NET_BANKING, WALLET
status	VARCHAR(10)	NOT NULL, CHECK	Transaction status: SUCCESS, FAILED, REFUNDED
transaction_ref	VARCHAR(30)	UNIQUE, NOT NULL	Unique banking gateway transaction reference number
paid_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp of financial transaction processing

Entity: 11. BAGGAGE — *Tracks checked and cabin baggage registered to specific passenger tickets.*

Relationships: • N : 1 with **ticket** (associated with passenger flight ticket)

Attribute Name	Data Type	Key / Constraint	Description & Rules
baggage_id	INT	**[PK] Auto Increment**	Unique surrogate primary key for the baggage piece
ticket_id	INT	**[FK] Ref: ticket**	Associated passenger flight ticket reference
baggage_type	VARCHAR(8)	NOT NULL, CHECK	Baggage classification: CABIN or CHECKED
weight_kg	DECIMAL(5,2)	NOT NULL, CHECK > 0 & <= 50	Weight of baggage piece in kilograms (up to 50 kg)
tag_number	VARCHAR(12)	UNIQUE, NOT NULL	Standard barcode baggage tag identification number

Entity: 12. EMPLOYEE — *Represents airline personnel, flight crews, ground staff, and hierarchical managers.*

Relationships: • N : 1 with **airline** (employed by airline)
• N : 1 with **airport** (stationed at base airport)
• N : 1 with **employee** (reports to manager; self-referencing hierarchy)

Attribute Name	Data Type	Key / Constraint	Description & Rules
employee_id	INT	**[PK] Auto Increment**	Unique surrogate primary key for the employee
airline_id	INT	**[FK] Ref: airline**	Employer airline company reference
base_airport_id	INT	**[FK] Ref: airport, NULLABLE**	Assigned base operating hub airport
manager_id	INT	**[FK] Ref: employee, NULLABLE**	Reporting supervisor / manager employee identifier
first_name	VARCHAR(50)	NOT NULL	Employee first given name
last_name	VARCHAR(50)	NOT NULL	Employee family / surname
email	VARCHAR(100)	UNIQUE, NOT NULL	Corporate communication email address
designation	VARCHAR(20)	NOT NULL, CHECK	Role: PILOT, CO_PILOT, CABIN_CREW, GROUND_STAFF, ENGINEER, MANAGER
hire_date	DATE	NOT NULL	Official joining and employment start date
salary	DECIMAL(10,2)	NOT NULL, CHECK > 0	Monthly compensation remuneration in INR